

15/7

1.8) Find the frequency counts for the following codes.

i) for $i = 1$; $i \leq n$; $i++$

printf ("Hello") ;

ii) $i = 1$;

$i = i + 20$;

printf ("%d", i) ;

iii) $i = 1$;

while ($i < n$).

{

printf ("%d", i) ;

$i++$;

}

2.8) for all the shell programs done in OS Lab, find the Time complexity in terms of Frequency count.

i)	Cost	Repetition	Total
$i = 1$	1	1	1
$i < n$	1	n	n
$i++$	1	$n-1$	$n-1$
$\text{printf}(\text{"Hello"})$	1	$n-1$	$n-1$
<u>Total :</u>			<u>$3n-1$</u>

ii)	Cost	Repetition	Total
$i = 1$	1	1	1
$i = i + 20$	1	1	1
printf	1	1	1
<u>Total :</u>			<u>3</u>

iii)	Cost	Repetition	Total
$i = 1$	1	1	1
$i < n$	1	n	n
printf	1	$n-1$	$n-1$
$i++$	1	$n-1$	$n-1$
<u>Total :</u>			<u>$3n-1$</u>

2) Sum of 2 No.s shell script

$\text{echo -n "Enter 1st no : "}$ → 1

read n1 → 1

$\text{echo -n "Enter 2nd no : "}$ → 1

read n2 → 1

sum = ϕ (n1 + n2) $\rightarrow 1$

echo "sum" $\rightarrow 1$

Total: 6